



COEP Technological University

(COEP Tech)
A Unitary Public University of Government of Maharashtra

Mid Semester Examination

Program : S.Y./T.Y. B.Tech.

Semester: III/V

Course Code: OEC-24007

Course Name: Linear Algebra

Branch:

Academic Year: 2025-26

Duration: 90 mins

Maximum Marks: 30

Date : 08/09/2025

Student MIS Number :

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Instructions :

1. Write your MIS Number and Branch on the Question Paper.
2. Writing anything else on the question paper is not allowed.
3. Read a given question very carefully and then answer.
4. Mobile phones and calculators are strictly prohibited.
5. Exchange/Sharing of stationery, etc. is not allowed.
6. Figures to the right indicate the course outcomes and full marks.
7. Unless otherwise mentioned, the field $F = \mathbb{R}$ for a given vector space V .
8. Let $M_{m \times n}(\mathbb{R})$ denote the vector space of matrices of order $m \times n$ with real number entries.
9. Any essential result, formula or theorem assumed for answering questions must be clearly stated.

1. Fill in the blanks **without justification**.

- (a) The additive inverse of $\begin{bmatrix} -1 & 0 & 1 & 3 \\ 7 & -2 & 3 & 0 \end{bmatrix}$ in $M_{2 \times 4}(\mathbb{R})$ is ... [CO2, 1 Mark]
- (b) Let $V = \left\{ f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is thrice differentiable, } \frac{d^2 f}{dx^2} + 25f = 0 \right\}$. Then V is a vector space of dimension ... over $\mathbb{R}$. [CO3, 1 Mark]
- (c) The zero element in $M_{2 \times 3}(\mathbb{R})$ is ... [CO2, 1 Mark]

2. State True or False **without justification**.

- (a) The set $\{(a, b, c) \mid a^2 = b^2\}$ is **not** a subspace of $\mathbb{R}^3$. [CO2, 1 Mark]
- (b) Consider the set $S = \mathbb{R}^2$, where addition is given by

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 y_2),$$

and scalar multiplication is given by $k(x_1, x_2) = (kx_1, x_2)$, for $k \in \mathbb{R}$. Then S is a vector space over $\mathbb{R}$ with respect to these operations. [CO3, 1 Mark]

- (c) The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = xy$ is a linear map. [CO2, 1 Mark]
3. (a) Let S be a non-empty subset of a vector space V . Define $\text{span}(S)$. [CO1, 1 Mark]
 (b) Does $\sin\left(\frac{\pi}{4} + x\right)$ belong to $\text{span}(\{\sin x, \cos x\})$? Justify. [CO2, 1 Mark]
 (c) Does $(2, -6, -3)$ belong to $\text{span}(\{(1, 2, 2), (-1, -1, 2), (1, -1, 1)\})$? Justify. [CO2, 2 Marks]
4. (a) Let W be the set of all skew-symmetric matrices in $M_{3 \times 3}(\mathbb{R})$. Prove that W is a subspace of $M_{3 \times 3}(\mathbb{R})$. [CO2, 2 Marks]
 (b) Find $\dim(W)$ with proper justification. [CO4, 2 Marks]
5. (a) Let u, v, w be three distinct vectors in a vector space V over $\mathbb{R}$. Show that $\{u, v, w\}$ is a linearly independent set in V if and only if $\{u + v + w, v + w, w\}$ is a linearly independent set in V . [CO4, 3 Marks]
 (b) Using the above or otherwise, check if $\{(2, 2, -3), (1, 2, -3), (1, 0, -3)\}$ is linearly independent in $\mathbb{R}^3$ or not. [CO3, 2 Marks]
6. (a) Define the range space and the null space of a linear map $T : V \rightarrow W$. Write down the rank-nullity Theorem. [CO1, (1+1) Marks]
 (b) Let $T_1 : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be given by

$$T_1(a, b, c, d) = (2b, 3c, 4d).$$

Show that T_1 is a linear map. [CO2, 2 Marks]

- (c) Find $\text{Im}(T_1)$, that is, the range space of T_1 . [CO3, 1 Mark]
 (d) Find $\ker(T_1)$, that is, the null space of T_1 . [CO3, 1 Mark]
 (e) Verify Rank Nullity Theorem for the linear map T_1 . [CO4, 1 Mark]
7. Let W_1, W_2 be two subspaces of V . Recall that

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\},$$

is a subspace of V . Let

$$W_1 = \{(x_1, x_2, x_3, x_4, x_5) \mid x_1 = x_2 - 5x_3 = 0\}, W_2 = \{(x_1, x_2, x_3, x_4, x_5) \mid x_1 = x_5 = 0\},$$

be two subspaces of $\mathbb{R}^5$.

- (a) Find $W_1 \cap W_2$. [CO5, 1 Mark]
 (b) Find $W_1 + W_2$. [CO5, 1 Mark]
 (c) Verify $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$. [CO5, 1 Mark]
 (d) Let V be a finite-dimensional space. Does

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$

hold? Only state true or false. [CO5, 1 Mark]
